Two-Sample t-test

Presentation Outline

Variables and Data

Variables

- One Quantitative Variable
- One Categorical Variable with 2 categories indicating sample/population membership

Data Collection

- One Sample containing Two Populations - Divide sample into two groups based on population membership
- Two Samples - one from each Population

Compare observations of quantitative variable between two populations

Parameters for Quantitative Variable

Population 1

- Mean: μ_1
- Variance: σ_1^2
- Std. Dev.: σ_1

Population 2

- Mean: μ_2
- Variance: σ_2^2
- Std. Dev.: σ_2

Data Table

Quantitative Variable	Categorical Variable
Y_{11}	Population 1
Y_{12}	Population 1
$\vdots$	$\vdots$
$\vdots$	$\vdots$
Y_{1n_1}	Population 1
Y_{21}	Population 2
Y_{22}	Population 2
$\vdots$	$\vdots$
$\vdots$	$\vdots$
Y_{2n_2}	Population 2

Summary Statistics

Sample 1

- Mean: $\overline{Y}_1 = \frac{\sum_{j=1}^{n_1} Y_{1j}}{n_1}$
- Variance: $S_1^2 = \frac{\sum_{j=1}^{n_1}(Y_{1j} - \overline{Y}_1)^2}{n_1 - 1}$
- Std. Dev.: $S_1 = \sqrt{\frac{\sum_{j=1}^{n_1}(Y_{1j} - \overline{Y}_1)^2}{n_1 - 1}}$

Sample 2

- Mean: $\overline{Y}_2 = \frac{\sum_{j=1}^{n_2} Y_{2j}}{n_2}$
- Variance: $S_2^2 = \frac{\sum_{j=1}^{n_2}(Y_{2j} - \overline{Y}_2)^2}{n_2 - 1}$
- Std. Dev.: $S_2 = \sqrt{\frac{\sum_{j=1}^{n_2}(Y_{2j} - \overline{Y}_2)^2}{n_2 - 1}}$

Null and Alternative Hypotheses

Null Hypothesis:

- $H_0 : \mu_1 - \mu_2 = 0$

Alternative Hypothesis:

- $H_a : \mu_1 - \mu_2 \neq 0$
- $H_a : \mu_1 - \mu_2 < 0$
- $H_a : \mu_1 - \mu_2 > 0$

Note: You can test for a value of $\mu_1 - \mu_2$ other than 0. However, no one does!

Sampling Distribution of $\overline{Y}_1 - \overline{Y}_2$

If H_0 is true, then

$$z = \frac{\overline{Y}_1 - \overline{Y}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

has a standard normal distribution for large sample sizes n_1 and n_2.

In order to use above result for inference, we would need to know σ_1^2 and σ_2^2.

σ_1^2 and σ_2^2 are population parameters and are generally unknown.

Equal Variance Assumption

Assume $\sigma_1^2 = \sigma_2^2 = \sigma^2$

With this assumption, if H_0 is true, we have

$$\frac{\overline{Y}_1 - \overline{Y}_2}{\sigma\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

has a standard normal distribution for large sample sizes n_1 and n_2.

Pooled Sample Variance

Estimate the unknown parameter σ^2 with S_p^2.

S_p^2 = pooled sample variance

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

$$= \frac{\sum_{j=1}^{n_1}(Y_{1j} - \overline{Y}_1)^2 + \sum_{j=1}^{n_2}(Y_{2j} - \overline{Y}_2)^2}{n_1 + n_2 - 2}$$

Two-Sample t-test Statistic

Replacing σ with S_p gives

$$t = \frac{\overline{Y}_1 - \overline{Y}_2}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

If H_0 is true, t has a t distribution with $n_1 + n_2 - 2$ degrees of freedom for large sample sizes n_1 and n_2.

p-value

Large values (in magnitude) of t indicate evidence against the null hypothesis.

Quantify this evidence with p-value $=$ probability of obtaining the test statistic t or a value more extreme (in direction of H_a) if H_0 is true.

- $H_a : \mu_1 - \mu_2 \neq 0$

  $p\text{-value} = 2 * P(t_{n_1+n_2-2} > |t|)$

- $H_a : \mu_1 - \mu_2 < 0$

  $p\text{-value} = P(t_{n_1+n_2-2} < t)$

- $H_a : \mu_1 - \mu_2 > 0$

  $p\text{-value} = P(t_{n_1+n_2-2} > t)$

Example - Hypothesis Test for $\mu_1 - \mu_2$

Creativity Data Sample Statistics

Group	Mean	Std. Dev.	Sample Size
Extrinsic	15.74	5.253	23
Intrinsic	19.88	4.440	24

$$
\begin{aligned}
S_p^2 &= \frac{(n_1 - 1) * S_1^2 + (n_2 - 1) * S_2^2}{n_1 + n_2 - 2} \\
&= \frac{22 * 5.253^2 + (23) * 4.440^2}{45} \\
&= 23.562
\end{aligned}
$$

Example - Hypothesis Test for $\mu_1 - \mu_2$

Null Hypothesis:

- Mean creativity score of the Extrinsic group is equal to the mean creativity score of the Intrinsic group
- $H_0 : \mu_1 - \mu_2 = 0$

Alternative Hypothesis:

- Mean creativity score of the Extrinsic group is different from the mean creativity score of the Intrinsic group
- $H_a : \mu_1 - \mu_2 \neq 0$

Example - Hypothesis Test for $\mu_1 - \mu_2$

Test Statistic

$$t = \frac{\overline{Y}_1 - \overline{Y}_2}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{15.74 - 19.88}{\sqrt{23.562}\sqrt{\frac{1}{23} + \frac{1}{24}}} = -2.926$$

p-value

$$2 * P(t_{45} > |-2.926|) = 2 * P(t_{45} > 2.926) = 0.0054$$

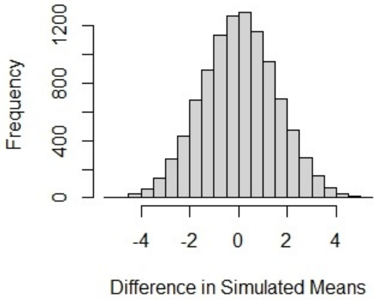
Example - Hypothesis Test for $\mu_1 - \mu_2$

Conclusion

We have very strong evidence the mean creativity score for the Extrinsic group is different from the mean creativity score for the Intrinsic group.

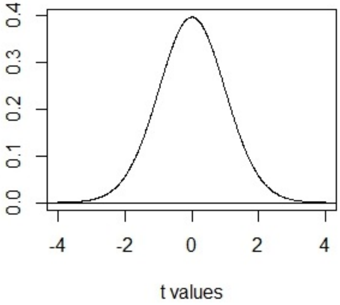
Randomization Test vs. Two-Sample t Test

Distribution of Difference in Simulated Means



p-value ≈ 0.0049

Sampling Distribution of Two Sample t-test statistic



p-value $= 0.0054$